

Abdelmalek Essadi University

Algebra IV — Final Exam

Year: 2022

Exercise 1:

Let $\alpha \in \mathbb{R}$. Consider the matrix $A_\alpha \in M_3(\mathbb{R})$ defined by:

$$A_\alpha = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ -\alpha & 1 & 1 \end{pmatrix}$$

1. Determine the **characteristic polynomial** of the matrix A_α .
2. For which values of α is the matrix A_α **diagonalizable** over $\mathbb{R}$?
3. Consider the equation in $M_3(\mathbb{R})$:

$$M^2 = A_\alpha \quad (*)$$

For which values of α does this equation $(*)$ possess **4 diagonalizable solutions** over $\mathbb{R}$?

Answer Area

1. Characteristic polynomial:

$$\det(A_\alpha - \lambda I) = \begin{vmatrix} 1-\lambda & 1 & 1 \\ 1 & 1-\lambda & 1 \\ -\alpha & 1 & 1-\lambda \end{vmatrix}$$

After simplification:

$$\chi_{A_\alpha}(\lambda) = -\lambda(\lambda - 2)(\lambda - \alpha)$$

2. Diagonalizability: A_α is diagonalizable if its eigenvalues are distinct: $\lambda = 0, 2, \alpha$. So $\alpha \neq 0$ and $\alpha \neq 2$.

3. Equation $M^2 = A_\alpha$: For 4 diagonalizable solutions over $\mathbb{R}$, A_α must be diagonalizable and have non-negative eigenvalues. Eigenvalues: $0, 2, \alpha \implies$ must be ≥ 0 . So $\alpha > 0$ and $\alpha \neq 2$ (to keep distinct eigenvalues). Hence $\alpha \in (0, 2) \cup (2, +\infty)$.

Exercise 2:

Consider the matrix $A \in M_3(\mathbb{R})$, defined by:

$$A = \begin{pmatrix} -1 & -2 & 0 \\ 2 & 3 & 0 \\ 2 & 2 & 1 \end{pmatrix}$$

1. Determine a **Jordan reduction relation** for the matrix A .
2. Deduce a **Jordan reduction relation** for the matrix $-A^{-1}$.
3. Determine the polynomials $\mathcal{Q} \in \mathbb{R}[X]$ such that the matrix $\mathcal{Q}(-A^{-1})$ is **diagonalizable**.

Answer Area

1. Jordan reduction for A : Eigenvalues: $\lambda_1 = 1$ (multiplicity 2), $\lambda_2 = 1$ (multiplicity 1). Jordan form:

$$J_A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

2. Jordan reduction for $-A^{-1}$: Eigenvalues: $\mu_i = -1/\lambda_i$. So $\mu_1 = -1$, $\mu_2 = -1$. Jordan form:

$$J_{-A^{-1}} = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

3. Polynomials $\mathcal{Q}$ for diagonalizability: $\mathcal{Q}(-A^{-1})$ is diagonalizable if it does not map the Jordan block to a non-diagonal one. This requires:

$$\mathcal{Q}'(-1) = 0$$

Hence, $\mathcal{Q}(X) = (X + 1)^2 \cdot R(X) + c$, where $R(X) \in \mathbb{R}[X]$ and $c \in \mathbb{R}$.

Exercise 3:

Let $\alpha \in \mathbb{R}$. Consider the matrix $A_\alpha \in M_3(\mathbb{R})$, defined by:

$$A_\alpha = \begin{pmatrix} 2\alpha + 2 & \alpha + 1 & 1 \\ -4\alpha - 2 & -2\alpha - 1 & -2 \\ 2\alpha - 1 & \alpha - 1 & 0 \end{pmatrix}$$

1. (a) $\text{Rg}(A_\alpha - I_3) = 2$
 (b) Characteristic polynomial $N_\alpha(X) = X(X - 1)(X + 1)$
 (c) $\dim E_{-1}(A_\alpha) = 1$
2. A_α is diagonalizable $\forall \alpha \neq 1$ (because all eigenvalues are distinct).
3. Minimal polynomial: $m_{A_\alpha}(X) = X(X - 1)(X + 1)$
4. $\alpha = 1$, $A = A_1$:
 (a) Jordan form: $J_A = \text{diag}(2, -1, 0)$
 (b) A and $P(A)$ are similar iff P permutes eigenvalues: $P(X) \in \mathbb{R}[X]$ s.t. $P(\{2, -1, 0\}) = \{2, -1, 0\}$

5. $\alpha = 0$, $A_0 = \begin{pmatrix} 2 & 1 & 1 \\ -2 & -1 & -2 \\ -1 & -1 & 0 \end{pmatrix}$:

- (a) Diagonalization relation: $A_0 = PDP^{-1}$, with $D = \text{diag}(1, -1, 0)$
- (b) $A_0^k = PD^kP^{-1}$
- (c) $M^2 = A_0 + I_3$ has solutions $M = P\text{diag}(\pm\sqrt{2}, \pm 1, \pm 0)P^{-1}$ with $\text{Tr}(M) > 0$

Answer Area

The complete Jordan and diagonalization computations follow standard linear algebra methods. All eigenvalues, minimal polynomials, and similarity conditions are provided above.